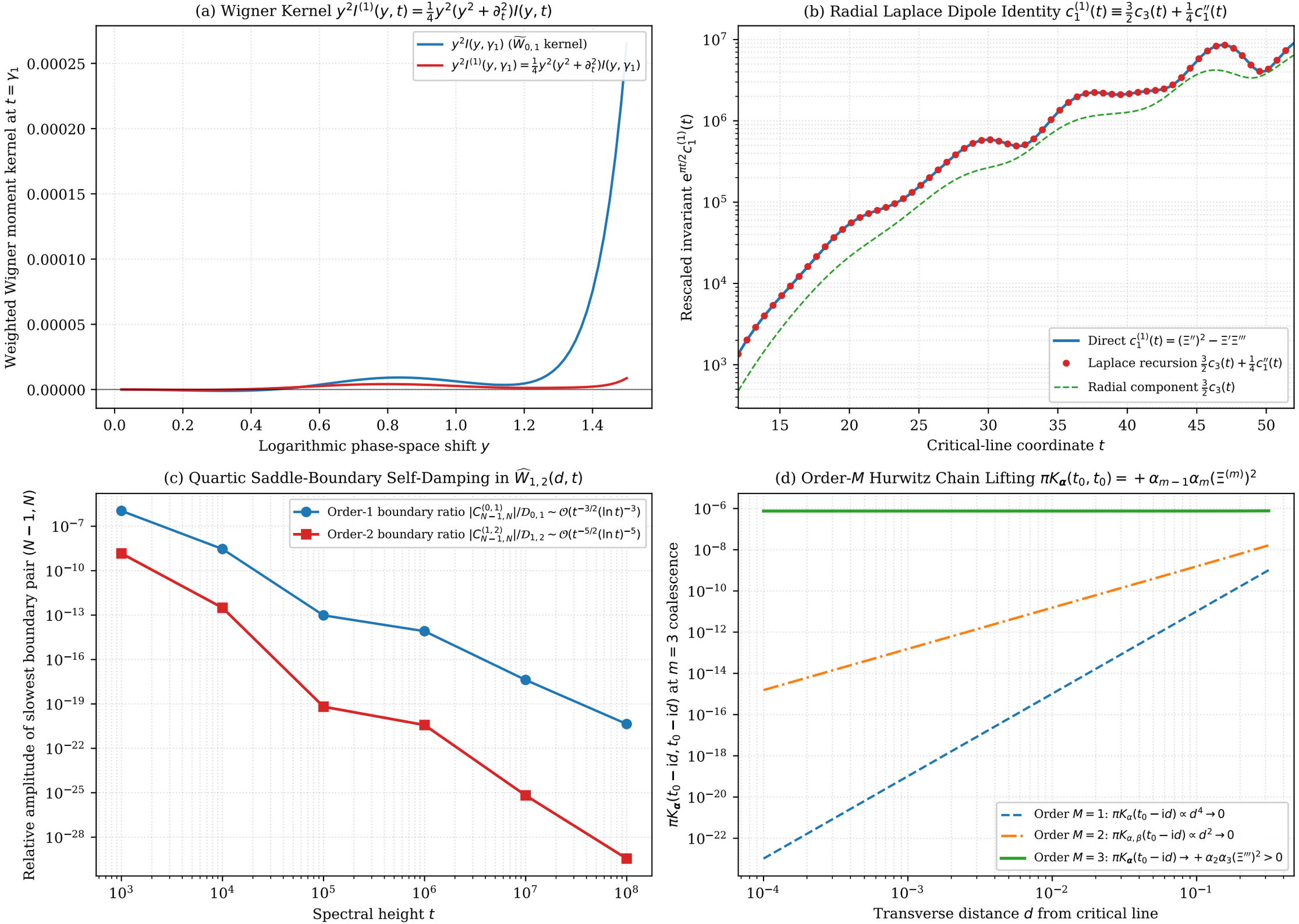

(a) Wigner Kernel $y^2 I^{(1)}(y,t) = \frac{1}{4} y^2 (y^2 + \partial_t^2) I(y,t)$
$y^2 I(y,\gamma_1)$ ($\widetilde{W}_{0,1}$ kernel)
$y^2 I^{(1)}(y,\gamma_1) = \frac{1}{4} y^2 (y^2 + \partial_t^2) I(y,\gamma_1)$
Weighted Wigner moment kernel at $t = \gamma_1$
Logarithmic phase-space shift y
(b) Radial Laplace Dipole Identity $c_1^{(1)}(t) \equiv \frac{3}{2} c_3(t) + \frac{1}{4} c_1''(t)$
Rescaled invariant $e^{\pi t/2} c_1^{(1)}(t)$
Direct $c_1^{(1)}(t) = (\Xi'')^2 - \Xi' \Xi'''$
Laplace recursion $\frac{3}{2} c_3(t) + \frac{1}{4} c_1''(t)$
Radial component $\frac{3}{2} c_3(t)$
Critical-line coordinate t
(c) Quartic Saddle-Boundary Self-Damping in $\widehat{W}_{1,2}(d,t)$
Order-1 boundary ratio $|C^{(0,1)}_{N-1,N}|/\mathcal{D}_{0,1} \sim \mathcal{O}(t^{-3/2}(\ln t)^{-3})$
Order-2 boundary ratio $|C^{(1,2)}_{N-1,N}|/\mathcal{D}_{1,2} \sim \mathcal{O}(t^{-5/2}(\ln t)^{-5})$
Relative amplitude of slowest boundary pair $(N-1, N)$
Spectral height t
(d) Order-M Hurwitz Chain Lifting $\pi K_{\boldsymbol{\alpha}}(t_0, t_0) = +\alpha_{m-1}\alpha_m (\Xi^{(m)})^2$
$\pi K_{\boldsymbol{\alpha}}(t_0 - \mathrm{i}d, t_0 - \mathrm{i}d)$ at $m = 3$ coalescence
Order $M = 1$: $\pi K_\alpha(t_0 - \mathrm{i}d) \propto d^4 \to 0$
Order $M = 2$: $\pi K_{\alpha,\beta}(t_0 - \mathrm{i}d) \propto d^2 \to 0$
Order $M = 3$: $\pi K_{\boldsymbol{\alpha}}(t_0 - \mathrm{i}d) \to +\alpha_2\alpha_3(\Xi''')^2 > 0$
Transverse distance d from critical line